

DAFFODIL INSTITUTE OF IT

B.Sc (Hons) in Computer Science & Engineering

Mid Term Examination of CSE 1st Year 1st Semester, 24th Batch, 2025

510207 (Physics)

Time – 1 hour 20 minutes, Full marks – 40

[N.B. – The figures in the right margin indicate full marks. Answer any two questions. All parts of the questions must be answered sequentially.]

- | | Marks |
|--|-------|
| 1. (a) State and explain Gauss's Law? | 4 |
| (b) Derive the expression of potential at a distance(r) from an isolated point charge (q) | 6 |
| (c) What is meant by Gaussian surface? Deduce Coulomb's law from Gauss's law. | 5 |
| (d) Two charges, $q_1 = +5\mu\text{C}$ and $q_2 = -3\mu\text{C}$, are placed 0.4 m apart in a vacuum. | 5 |
| I. Find the electrostatic force between them. | |
| II. Find the electric field intensity at a point P, which is 0.2 m from q_1 along the line joining the two charges. | |
| III. Find the electric potential at point P due to both charges. | |
| 2. (a) Define electric dipole, dipole moment, quantization of charge | 4 |
| (b) Find the expression for electric field intensity at a point on the perpendicular bisector of a dipole. | 6 |
| (c) Derive an expression for electric potential due to a charged disk. | 5 |
| (d) Expression for Torque and potential energy of an electric dipole in a uniform electric field. | 5 |
| 3. (a) What is capacitance? Derive an expression for the energy stored in a charge capacitor. | 6 |
| (b) State and explain Kirchhoff's Voltage law. | 4 |
| (c) What is parallel plate capacitor? Calculate capacitance for parallel plate capacitor? | 5 |
| (d) A rectangular carbon block has dimensions of $1.0\text{ cm} \times 1.0\text{ cm} \times 50\text{ cm}$. | 5 |
| I. What is the resistance measured between the two square ends? | |
| II. What is the resistance measured between two opposing rectangular faces? The resistivity of carbon at 20°C is $3.5 \times 10^{-8}\text{ ohm-meter}$. | |

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- | | | Marks |
|----|--|-------|
| 1. | (a) What is Coulomb's law? Deduce Coulomb's law from Gauss's law | 4 |
| | (b) What is meant by electric field intensity? Write the expression for electric field intensity due to point charge? | 6 |
| | (c) An infinite length rod have linear charge density λ . Calculate E for the rod at a distance 'r' from the rod. | 5 |
| | (d) Two point charges, $q_1 = +3\mu\text{C}$ and $q_2 = -2\mu\text{C}$, are placed 0.5 m apart in vacuum. Find the magnitude force, electric field intensity (E) and direction of the electrostatic force between them. | 5 |
| 2. | (a) What is electric dipole? Calculate electric potential for an electric dipole. | 4 |
| | (b) Write the expression of electric field intensity due to a ring of charge q and of radius a. | 6 |
| | (c) Deduce an expression of electric field intensity due to a long straight, uniformly Charged wire. | 5 |
| | (d) Define dielectric constant? What is three electric vector? | 5 |
| 3. | (a) Calculate equivalent capacitance for series and parallel capacitor, Where $c_1 = 5\mu\text{F}$, $c_2 = 8\mu\text{F}$, $c_3 = 10\mu\text{F}$? | 5 |
| | (b) State and explain Kirchhoff's Current law. | 5 |
| | (c) Define the terms current, current density, resistance and resistivity, electric potential? | 6 |
| | (d) Two charges, $q_1 = +5\mu\text{C}$ and $q_2 = -3\mu\text{C}$, are placed 0.4m apart in a vacuum. Find the electric potential at point P due to both charges and find total electric potential at point. | 4 |

(27/2/24)

$$F = 46\text{E}$$

$$\frac{1}{\mu}$$

